

Max Lee | EIT

mtxlee12@gmail.com | (949) 771-4477 | [in](https://www.linkedin.com/in/maxtlee) : linkedin.com/in/maxtlee | Personal Website: maxtylee.github.io/portfolio

EDUCATION

University of California, Berkeley

August 2024 - May 2025

Master of Engineering, Mechanical Engineering with a specialization in Control of Robotic and Autonomous Systems

University of California, San Diego

September 2020 - June 2024

Bachelor of Science, Mechanical Engineering with a specialization in Controls & Robotics

SKILLS

3D CAD Software & Prototyping: SolidWorks, Inventor, Onshape, GD&T, 3D Printing (FDM/SLA), CNC, DFM/DFA

Hardware Development: Machine Tools (Lathe, Manual Mill, Band Saw), Rapid Prototyping, Laser Cutting, Soldering

Software & Programming: Python, MATLAB, ROS/ROS 2, OpenCV, ANSYS Simulation (FEA/CFD), Microsoft Office

RELEVANT EXPERIENCE

LUXE Bidet

October 2025 - Present

Mechanical Design Engineer

- Drive mechanical product development from concept to mass production, owning CAD design, prototyping, design validation, and manufacturing coordination while optimizing designs for manufacturability, quality, and scalability.

YQN Logistics Co., Ltd.

July 2025 - October 2025

Mechanical Engineering Intern

- Led a team of 30 to deploy 12,000+ rack-mounted LED pick-to-light indicators, completing implementation 4 days ahead, enabling earlier adoption of a system that improved picking efficiency and reduced cost per light by 5%.
- Developed SOPs and Value Stream Maps (VSM) tailored to the robotic sorting system, aligning workflows with mechanical equipment requirements to ensure smooth operation and consistent performance.

BurnBot, Inc.

September 2024 - May 2025

Mechanical Engineering Project Lead – Capstone Project

- Led development of a mechanical obstacle detection system achieving **less than 0.01 inches** of measurement error, directly supporting BurnBot's autonomous wildfire mitigation vehicle.
- Validated blade performance through physical prototyping of polycarbonate, stainless steel, and spring steel variants, selecting **0.042 inch** spring steel as the optimal design for repeatable deflection and reliable sensing.
- Developed a full-scale Onshape CAD assembly of the final system with integration-ready mounting, enabling direct transfer to BurnBot's manufacturing team.
- Built two mock torch assemblies and a pulley-driven demo rig for expo day, demonstrating the system to 100+ attendees and industry professionals.

Neubo Inc.

February 2024 - June 2024

Mechanical Design Lead – Capstone Project

- Designed and fabricated pneumatic NOT gates using acrylic lamination, achieving reliable signal inversion within an 8–12 psi operating range and integrating them into NAND, NOR, OR, and AND circuits.
- Fabricated a silicone soft robotic gripper using silicone molding techniques capable of lifting at least **2 lb** and withstanding pressures up to **10 psi**, ensuring safe operation.
- Developed **10+** unique pneumatic logic gate circuits for engineering education, supported by **technical documentation** outlining circuit implementation and manufacturing processes.

Extremum Seeking Control Research Lab

September 2023 - June 2024

Undergraduate Research Assistant – Professor Miroslav Krstic & Mamadou Diagne

- Implemented ROS 2-based autonomous control architecture using Python, integrating Raspberry Pi sensors and actuators with an extremum-seeking optimization algorithm.

OTHER PROJECTS

- **Ping-Pong Playing Robot:** Calibrated a four-camera vision system using Python and OpenCV to derive transformation matrices for accurate 3D ball localization, achieving a **0.04** reprojection error. Automated checkerboard image selection and matching, reducing calibration time by **75%**.
- **Wearable Motion Tracker:** Designed and 3D printed an IMU-based wrist tracker in SolidWorks using DFMA principles, then validated motion data through user testing and Euler angle analysis for real-time tennis feedback.
- **Autonomous Vehicle Navigation System:** Integrated the Google Maps Directions API with the Donkey Car framework to enable autonomous route planning, automatically triggering GPS-based navigation to charging stations upon low-battery detection **with 1.5 m** positional accuracy.